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CENTRE
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CANDIDATE
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CHEMISTRY

Grade 12

Paper 3

May 2024

1 hour 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Calculator
Qualitative Analysis Notes and Periodic Table
Ruler

12CHEM/03

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 30.

Answer all questions in English.

For Examiner's Use	
1	
2	
Total	

This document consists of **8** printed pages.



- 1 A student determines the percentage by mass of iron(II) ions in an iron(II) salt by titration with acidified potassium manganate(VII), KMnO_4 , using the following steps.

Step 1 Accurately weighs 4.50 g of the iron(II) salt and pour it into a 250 cm^3 beaker.

Step 2 Adds distilled water to the beaker until all the unknown iron(II) salt dissolves.

Step 3 Transfers this solution into the 250 cm^3 graduated flask, fills it with water, and labels it as **B**.

Step 4 Transfers 25.0 cm^3 of solution **B** into a conical flask.

Step 5 Adds 10 cm^3 of 1.00 mol dm^{-3} sulfuric acid to the conical flask.

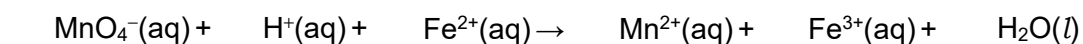
Step 6 Fills a burette with the 0.0200 mol dm^{-3} solution $\text{KMnO}_4(\text{aq})$.

Step 7 Records the initial burette reading.

Step 8 Titrates solution **B** against the $\text{KMnO}_4(\text{aq})$ until the solution changes to a permanent pink colour.

Step 9 Records the final burette reading.

- (a) (i) Complete the redox equation.



[1]

- (ii) Use changes in oxidation states to identify the reducing agent and the oxidising agent in this reaction.

reducing agent

oxidising agent

[2]

- (b) The student obtains the results shown in the table.

- (i) Complete the table and state which titrations you use to calculate the mean titre.

Calculate the mean titre without including the rough titration.

titration number	1	2	3	4
initial (burette) reading/ cm^3	0.0	1.0	1.0	0.5
final (burette) reading/ cm^3	27.8	28.0	27.9	27.9
volume of solution B added/ cm^3				

titrations used

mean titre = cm^3

[3]

(ii) Use the titration results to calculate the number of moles of MnO_4^- .

..... mol [1]

(iii) Calculate the number of moles of iron(II) salt in 250 cm^3 of solution.

..... mol [2]

(iv) Assume that in 1 mole of the iron(II) salt contains 1 mole of iron(II) ions.
Calculate the mass of Fe^{2+} ions in iron(II) salt.

..... g [1]

(v) Calculate the percentage of mass of Fe^{2+} ions from the total mass of iron(II) salt.

..... % [1]

(c) The student decides that the change in colour during the titration is the equivalence point.

(i) State and explain whether the student is correct.

.....
..... [1]

(ii) The error in a single burette reading is $\pm 0.05 \text{ cm}^3$.
What is the percentage error in the titre volume calculated in (b)(i)? Give your answer to 2 significant figures.

percentage error = % [1]

(iii) The student uses a measuring cylinder to transfer solution **B** to the conical flask.
Suggest a more accurate piece of apparatus to use instead of a measuring cylinder.

..... [1]

[Total: 14]

- 2 The enthalpy change, ΔH_r , for the decomposition of calcium carbonate to calcium oxide cannot be measured directly.



The student determines the enthalpy change for the reactions of calcium carbonate and calcium oxide with hydrochloric acid. Excess hydrochloric acid will be used in both experiments.

Then the student uses Hess' Law to calculate the enthalpy change for the reaction above.

- (a) State Hess' Law.

.....
 [1]

- (b) The student carries out two separate experiments and uses the results of these experiments to determine the enthalpy change of reaction for the decomposition of CaCO_3 by following these instructions.

Experiment 1 with $\text{CaCO}_3(\text{s})$ and $\text{HCl}(\text{aq})$

Step 1 Uses a measuring cylinder to place 50 cm^3 of 3.0 mol dm^{-3} hydrochloric acid into a 250 cm^3 glass beaker.

Step 2 Records the temperature of the hydrochloric acid.

Step 3 Places 2.5 g of calcium carbonate into the hydrochloric acid.

Step 4 Stirs the mixture using a thermometer and records the highest temperature obtained.

Step 5 Records the results in a table.

Student obtains the following results from Experiment 1.

- (i) Complete the table by calculating the temperature change.

Experiment	$\text{CaCO}_3(\text{s})$
temperature of acid at start / $^{\circ}\text{C}$	24.5
highest temperature of reaction mixture / $^{\circ}\text{C}$	28.0
temperature change / $^{\circ}\text{C}$	

[1]

- (ii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixture has a density of 1.0 g cm^3 .

Use the temperature change in **(b)(i)** to calculate the heat released in Experiment 1.

heat released = J [1]

- (iii) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_1 .



enthalpy change = kJ mol^{-1} [2]

(c) Experiment 2 with $\text{CaO}(\text{s})$ and $\text{HCl}(\text{aq})$

The student:

Step 1 Uses a measuring cylinder to place 50 cm^3 of 3.0 mol dm^{-3} hydrochloric acid into a 250 cm^3 glass beaker.

Step 2 Records the temperature of the hydrochloric acid every minute for a total of eight minutes.

Step 3 Records the results in the table.

Step 4 At exactly 2.5 minutes adds 1.4 g of the calcium oxide into the hydrochloric acid.

time/minutes	0	1	2		3	4	5	6	7	8
temperature of mixture/ $^\circ\text{C}$	24.0	24.5	24.5		35.0	37.5	36.0	34.5	35.0	31.5

- (i) Use the results from Experiment 2 and plot a graph of temperature against time on the grid below.

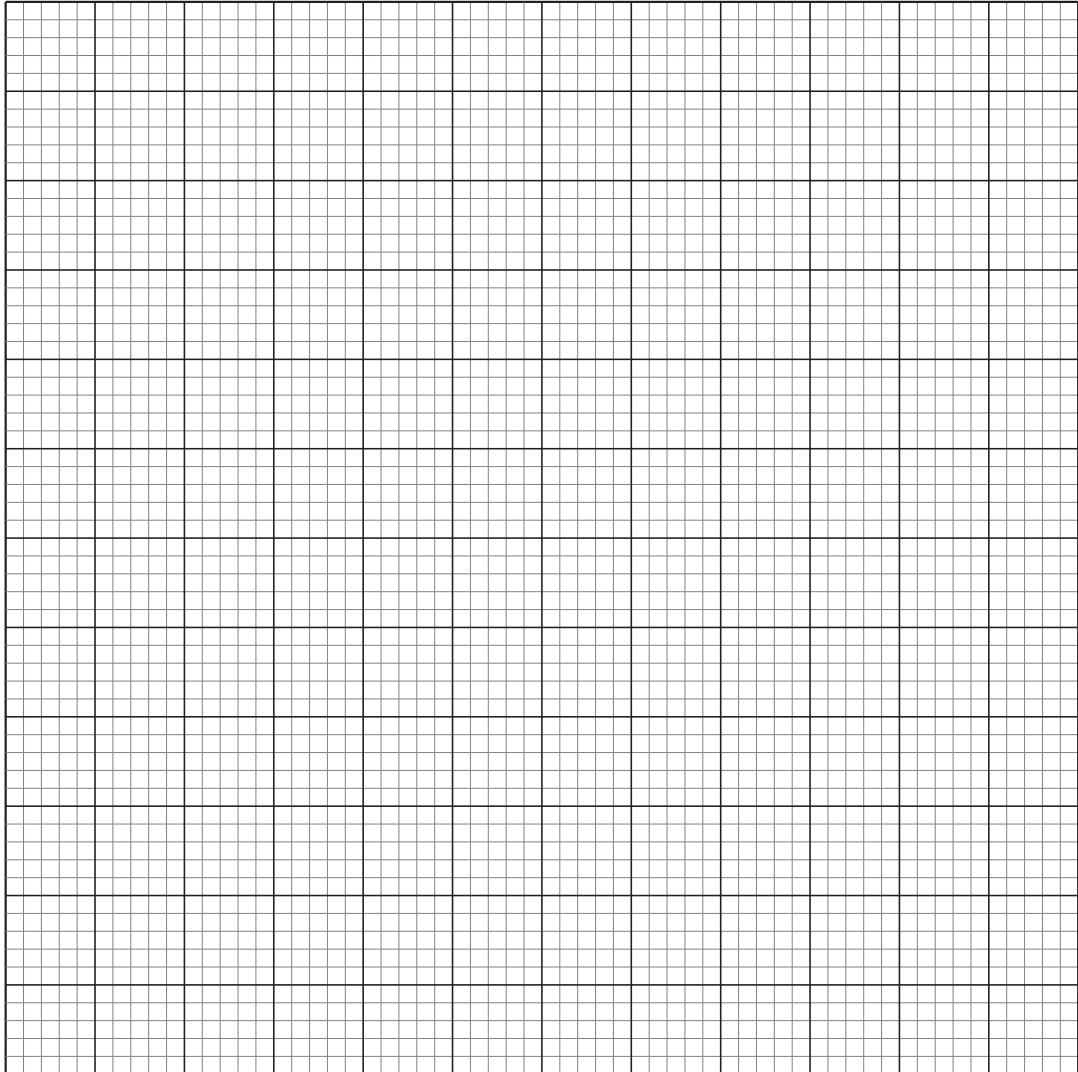
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Draw two lines of best fit for Experiment 2 through the points on your graph.

The first line is for the first three temperatures during the first two minutes.

The second line is for the last six temperatures during the last five minutes.

Extrapolate the lines to estimate the temperatures at 2.5 minutes and the temperature change, ΔT , at 2.5 minutes.



Calculate the maximum temperature change at 2.5 minutes.

$$\Delta T = \text{.....}^{\circ}\text{C} \quad [4]$$

- (ii) Circle the most anomalous point on your graph and suggest a reason for the anomalous result.

.....

..... [1]

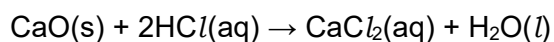
- (iii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixtures has a density of 1.0 g cm^3 .

For
Examiner's
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Use the temperature change in (c)(i) to calculate the heat released in Experiment 2.

heat released = J [1]

- (iv) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_2 .



enthalpy change = kJ mol^{-1} [1]

- (d) (i) Use the equations in parts (b)(iii) and (c)(iv) to construct the Hess' Law cycle to show how you could calculate the enthalpy change for the decomposition of calcium carbonate, ΔH reaction.
Label the arrows in your cycle.

[2]

- (ii) Use your values for enthalpy change from Experiment 1 and Experiment 2 to calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r

For
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ΔH_r kJ mol⁻¹ [1]

- (e) The procedure in Experiment 2 is repeated using the same mass of calcium oxide.

This time 50 cm³ of 4.0 mol dm⁻³ HCl is used instead of 50 cm³ of 3.0 mol dm⁻³ HCl.

Compare the maximum temperature change in this case with the temperature change in (c)(i).

.....
..... [1]

[Total: 16]

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